State Methodology (10/20)

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This memo describes the state extension of the Budget Lab's ECEC model from national survey data to local care demand and equilibrium prices.

Processing and calibration run nationally for the 2019 base year. State simulations cover children ages 0–4 in 50 states, DC, and New York City (NYC). I exclude NYC's separate run from national totals to avoid double counting with New York State.

Methods follow the national model except where noted.

The state extension separates:

- National estimates and assumptions: the care, employment, and price-wedge forests; behavioral parameters β and ρ; and supply parameters.
- Imposed for each geography: annual care hours in six care types, base wages, staffing rules, non-labor costs, and the employment path. Each year, I set the baseline employment rate of each caregiver group (sex by four age bands) to its projected rate.
- Predicted by the model: the split between low- and high-price centers, equilibrium prices, equilibrium wages, policy-induced employment changes, and other responses to policy.

1 National foundation

The care model learns from the 2019 National Survey of Early Care and Education (NSECE). The simulator's households come from the 2019 American Community Survey (ACS). I add state and county identifiers to the ACS households.

Each child gets one care label. The label follows the national model's provider-priority rules, from Borowsky et al. (2022). Paid centers take priority over unpaid centers. The model has seven non-parental care types, each part-time or full-time, plus Parent Only.

The national model ranks families on nominal income. For state runs, I adjust income for local prices. I multiply income by 100 and divide by the 2019 BEA all-items regional price parity. I then apply national cutpoints built separately in NSECE and ACS.

ACS households use their state's parity. NSECE identifies only the Census region. Thus NSECE households use the child-weighted mean parity of the states in their region. NYC uses the New York metro parity, 114.9. NYC gets its own ranks and alpha.

National forests predict care for a given caregiver employment status. The forests keep the joint care probabilities for the two youngest children. I combine these with employment probabilities. The result is the baseline choice probabilities, p_0.

I keep the national utility specification and calibration targets. The targets are -0.15 for employment with respect to care cost and -0.05 for work hours with respect to income. The price-adjusted calibration gives utility scale $\beta = 0.0003405$ and curvature $\rho = 0.0262$. Both elasticities hit their targets.

Alpha is a preference constant for each family and choice that reproduces p_0. State runs adjust alpha as described below. A family's care cost is its price wedge times the sector price times its annual care hours. The **price wedge**, predicted from NSECE, is the ratio of a family's payment per hour to its sector's hourly price.

2 Center price tiers

The national model separates low- and high-price household center care at $4.75 an hour. For state runs, I classify posted prices and household payments with a threshold. The threshold changes by region, urban density, and age band. A price at or above the threshold is high-price. The tiers use price as a proxy for quality; quality is not directly observed.

The regional samples for rural and moderate density are small. I start with each region's high-density median. I scale it by the national price ratio between density groups:

$$\begin{array}{c}\text{threshold}\\(\text{region, urban density, age band})\end{array} = \begin{array}{c}\text{regional median price,}\\\text{high urban density, age band}\end{array} \times \frac{\begin{array}{c}\text{national median price,}\\\text{this urban density, age band}\end{array}}{\begin{array}{c}\text{national median price,}\\\text{high urban density, age band}\end{array}} \tag{1}$$

I calculate medians over positive prices. The weight for each center is its survey weight times its full-time-equivalent enrollment. The posted price is the weekly rate divided by 40 hours. A household's price is its weekly payment divided by its weekly hours. The age bands are 0–1, 2–3, and 4–5.

Urban density is the NSECE community urban density. It has three levels:

- High: 85% or more of the nearby population is urban.
- Moderate: 30% to under 85% is urban.
- Rural: under 30% is urban.

A center uses its own level. A family uses the level of its household. Each regional cell has 5–35 surveyed rural centers and 28–134 moderate-density centers per age band (unweighted counts of centers). I assume that the national ratio holds in every region. The care forest has no urban density feature.

3 Demand targets for each geography

A state's households have national preferences. I re-anchor each geography to its own level of care. The target is annual hours by care type. The national model has no local demand targets.

3.1 Children in regular care

$$\begin{array}{c}\text{children in}\\\text{regular care}\end{array} = \begin{array}{c}\text{population}\\\text{ages 0–4}\end{array} \times \text{NSCH rate} \times \begin{array}{c}\text{five-hour}\\\text{factor}\end{array} \tag{2}$$

The population is the Census 2019 county estimate for ages 0–4.

The National Survey of Children's Health (NSCH) rate is the share of children with at least 10 weekly hours of non-parental care. I pool the rate over 2016–2019 in each state. NSCH does not identify NYC. NYC uses New York's rate.

Regular care in NSECE starts at 5 hours a week, not 10. I scale the rate by one national five-hour factor. The factor is NSECE children in regular care divided by NSECE children in care 10 or more hours a week. Both counts use NSECE child weights. Both counts cover children ages 0–4 and exclude school programs. The factor is 1.098. The numerator has 2,814 surveyed children (unweighted).

For the 50 states and DC, the result is 11.6 million children in regular care (weighted). This is 59.2% of the estimated population of 19.6 million children ages 0–4.

3.2 Unpaid center children

Unpaid Center-Based care takes each geography's own program counts:

1. Pre-K is the larger of public-school pre-K from the Common Core of Data and an all-settings count. The all-settings count is Table 2 of the NIEER State of Preschool yearbook for a state, and the city's own count for NYC.

2. Funded Head Start and Early Head Start slots convert to enrolled children with the Office of Head Start enrollment ratios from the 2018–19 Program Information Report.

3. I remove home-based and family child care options, children age 5 and older, and Head Start children that pre-K already counts. I use one pooled Head Start overlap share: the slot-weighted mean across the 18 geographies with NIEER estimates.

I count adjusted pre-K and Head Start enrollment as unpaid-center children. This count can be too high, for two reasons. Children who also use paid centers get the paid-center label. Some public programs charge families. The overlap adjustment removes children that both pre-K and Head Start count. It does not remove overlap with other care.

3.3 Other care types

I subtract unpaid-center children from total children in regular care. I divide the remainder across the other five types with child-weighted p_0 shares in each geography. The same probabilities give the part-time share of each type. The mix shows local household characteristics under the national model.

3.4 Hours and re-anchoring

$$\text{annual hours of type} = \text{children in type} \times \left(\text{part-time share} \times 500 + \text{full-time share} \times 2{,}000\right) \quad (3)$$

I target six categories: paid center, unpaid center, paid home, unpaid home, other paid, and other unpaid care. The paid-center target combines low- and high-price centers. Part-time and full-time care represent 10 and 40 hours a week for 50 weeks.

At simulation start, I adjust alpha in two steps:

1. I change the preference constants at the geography's 2019 supply prices. Local prices alone do not change the initial choice probabilities, p_0.

2. I solve one constant δ for each care type. Every household in the geography shares the same δ. The δ values make base-year hours match the file. Parent Only has $\delta = 0$. A household's alpha for a choice moves by the sum of δ over the care types of its children. The two center tiers share one δ. The model sets the split between them.

In the same solve, I add one shift on working choices for each caregiver group. The groups are sex by four age bands. The shift makes each group's 2019 employment rate equal its employment-forest rate. The run stops if the targets lack support or the solve does not converge.

The joint solve targets care hours and employment rates; child counts and part-time shares remain model outcomes.

Matching care and employment jointly aligns base-year care demand with the employment rates from which yearly targets grow.

At initialization, I check that rebuilt net income matches calibration values where applicable, that repriced alpha recovers p_0, and that sector hours at local prices match the re-anchored hours. Validation of the price-adjusted joint solve across all 52 geographies is pending.

4 Supply and employment for each geography

Each sector uses fixed amounts of labor from workers with and without a bachelor's degree. Total demand across sectors sets each group's wage. A paid sector's price equals its labor cost plus its non-labor residual. Unpaid centers charge households zero, but they add to labor demand. The model does not track their financing.

I set each group's base-year labor supply equal to the labor required for re-anchored care hours. Each geography's supply file changes three national inputs: wages, staffing, and the residual.

4.1 Wages

I scale each group's national base wage by its state-to-national wage index ratio. Workers without and with a bachelor's degree have national base wages of $12.06 and $19.70 an hour. The wage index mixes May 2019 hourly medians from the BLS Occupational Employment Statistics for childcare workers, preschool teachers, and preschool and childcare administrators. The mix uses the national ACS share of each group's workers in each occupation. NYC uses the New York metro area.

4.2 Staffing

The staffing index is the state's required staff per child divided by the national average. The national average weights states by children ages 0–4. The rules come from the 2017 Child Care Licensing Study (ICPSR 37700).

For centers, required staff per child is the average of 1 over the child-to-staff ratio at five ages: 11 months, 18 months, 35 months, 3 years, and 4 years. NSECE 2019 center enrollment at those ages gives the weights. For homes, required staff per child is 1 over the maximum number of children in a family home. A state with no family-home licensing ratio uses an index of 1. NYC centers use the June 2019 city Health Code. Idaho's center ratios come from its statutory points rule.

I multiply national labor requirements per child-hour by the center index for all three center sectors. I multiply them by the home index for paid homes. Each index applies equally to both worker groups. Classrooms have more staff than the limits require. I use the licensing limits because they provide a consistent basis for geographic comparisons, with a fallback for unlicensed family homes.

4.3 Non-labor costs

The national model uses one national residual per sector: the hourly price minus labor cost per child-hour. These are $0.51 for low-price centers, $4.61 for high-price centers, and $2.83 for paid

homes.

I scale each sector's national non-labor residual by the geography's 2019 BEA housing-services price parity divided by 100. NYC uses the metro value. Unpaid centers keep a zero residual. The factor goes from 0.55 to 1.70 across the 52 geographies. The residual covers non-labor costs and provider margins. I use housing-services prices as a proxy for how non-labor costs change from place to place.

I compare model prices with the National Database of Childcare Prices (NDCP) in 46 states. I compare two versions of the model price: one with the residual scaled by the housing parity, and one with the national residual, as in the national model. The housing scaling raises the unweighted Pearson correlation from 0.83 to 0.89. It lowers the weekly root mean squared error from \$41.6 to \$34.1. The ratio of the child-weighted mean model price to the child-weighted mean NDCP price rises from 1.09 to 1.12.

The comparison uses 2019 base supply prices converted to 40-hour weeks, with fixed model enrollment weights across price tiers and age groups. NDCP prices are means of county medians, weighted by children ages 0–4 and including imputed prices. The sources' age bands and full-time definitions do not exactly align:

- Age bands. NDCP infant, toddler, and preschool prices cover 0–23, 24–35, and 36–54 months. I match them to model ages 0–1, 2, and 3–4. Model age 4 includes children aged 55–59 months, which NDCP does not price.
- Full time. The model week is 40 hours. NDCP has no single full-time definition: each state's market rate survey sets its own, and some states let providers decide. NDCP converts hourly rates to weekly at 40 hours and daily rates at 5 days.

4.4 Employment path

A geography's projected employment rate, by sex and age band, is the national projected rate times the state-to-national ACS rate ratio. The national series uses Current Population Survey (CPS) history. I project it forward with CBO's long-term labor force participation projections. The ratios use ACS two-year pools. After 2024, the ratios stay at their 2024 values.

I aggregate the path to four caregiver age groups by sex. Then I apply the growth since 2019 to each group's base-year employment rate. The base-year rate is the weighted mean of the group's employment-forest probabilities.

5 Simulation

Each year, I update household weights, earnings, income, and prices using national growth factors. Each geography is its own market. Workers and families do not move between geographies.

The solver starts from the prior year's prices. Group-specific utility shifts match baseline employment targets and stay fixed across policy scenarios. Baseline and policy runs use the same Gumbel shocks. Supply links demand to price through labor:

$$\text{wage of group} = \text{base wage} \times \text{wage growth} \times \left(\frac{\text{labor demand}}{\text{base-year labor}} \right)^{1/\text{supply elasticity}} \tag{4}$$

The supply elasticities are 4 for workers without a degree and 2 for workers with a degree. These are assumed values from Borowsky et al. (2022). I use the national values in every geography. The residual grows with the CPI.

6 Open questions

1. Should state base-year targets sum to national care hours, or should I report the discrepancy?
2. Should local prices change baseline care probabilities, or should local alpha absorb their effect before demand targeting?
3. Is regional price adjustment in NSECE close enough to state adjustment in ACS? The forests learn from regional parities but apply state parities.
4. What are the empirical sources for the -0.15 and -0.05 calibration targets?